

Opening remarks

What the application does

Course discovery
Evaluations and attribution
Profiles, memories, and history
Schedule audits
Schedules, calendar, and chat

How the application works

Logical Layers
Components and integrations
Deployment view
Chat request and SSE
Unified search pipeline
Mutations and clarification
Parallel Audit and Quality Gate
Output validation (civility, policy)
Agent Overview
Automated tests, evaluation, and observability

Closing remarks

Limitations
Future work and reflection
Questions

Atlas

AI-Assisted Schedule Builder for JHU Undergraduates

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Johns Hopkins University
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Opening: Problem, Users, Why AI

Problem

JHU students often struggle with the *course selection process*, specifically, there are many struggling issues:

1. The students need to discover courses, respect constraints, and build feasible schedules
2. The advisors are not entirely accessible to students all the time to provide advises all the time
3. The advisors would not be capable of providing comprehensive career information or evaluations of instructors, while these information are often scattered around the internet

Targeted Users

Undergraduates planning a term: search, shortlist, compare workload, and iterate with advising-style help.

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Opening: How AI can Help

The advisors won't be available 24-7, and if one is consulting information on the internet, it will take significant time, and regular AI models are not tuned with your preferences and information.

Why AI is essential (not bolted-on)

- ▶ **Discovery:** Semantic search + match explanations over embeddings; an agent orchestrates tools instead of static forms.
- ▶ **Personalization:** Onboarding → structured memories; schedule-aware chat; natural-language add/drop/swap.
- ▶ **Reliability:** Clarification before risky writes; unified search; parallel audits with quality gates on outputs.

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Demo 1 — Course discovery

Goal

Open-ended query → merged ranked results from unified searchCourses with explicit matchType and readable matchExplanation; clarification when the request is ambiguous.

- ▶ Live: NL query → cards with matchType (exact / constraint / semantic / hybrid) and substantive explanations, including backfill when the agent forces a narrow search payload.
- ▶ Optional path: ambiguous intent → disambiguation UI with saved multi-turn state.
- ▶ When the query names a **professor** or **specific course**, parallel **RateMyProfessor** + **Reddit** lookups can augment internal evals; tool use stays scoped so generic browse queries stay lightweight.



Demo 2 — Eval summaries + attribution

Goal

On-demand eval summary grounded in metrics with an inspectable source table; course metrics default to **cross-term aggregation** unless the user or tool pins a specific term.

- ▶ Show: “Summarize evals” on a course card → narrative + numeric claims tied to evidence.
- ▶ Open **raw** / **source** modal: metric names, values, terms, instructors; each claim traceable to a row in the extended attribution view.
- ▶ If no backing eval rows: hide raw access and do not fabricate a summary.

Demo 3 — Onboarding, memories, history

Goal

Onboarding and chat produce structured memories; completed-course history and transcript intake feed recommendations, prerequisite logic, and audits.

- ▶ Show: onboarding or profile → `derived_memories` / memory list with phrasing tuned for downstream agent use.
- ▶ Completed courses: add/remove from history; optional **transcript** paste or upload → review list with `matched` / `ambiguous` / `unmatched` before anything is saved.
- ▶ Keep one short path showing how history drives **prerequisite readiness** and exclusions in planning or audit surfaces.



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Demo 4 — Audit (parallel checks + synthesis)

Goal

From the schedule context, run a **parallelized** audit that merges normalized checks into one report: workload, schedule conflicts, preference alignment for days and times, and prerequisite readiness, then apply an audit-only **quality gate** validate → one regenerate → fallback, with **partial results** when an individual check fails.

- ▶ Walk findings by category with severity and evidence, including **preference alignment** against stated days and times.
- ▶ Call out **prerequisites** and **conflicts** alongside workload so plan fit is grounded in concrete checks, not a separate opaque score.
- ▶ Note the async **offline evaluator** on completed audit sends for observability; be ready in Q&A to contrast audit vs. non-audit output policies if asked.

Demo 5 — Schedules, calendar, and schedule-aware chat

Goal

From the dashboard, open a schedule, use the **list vs. weekly calendar** view, optionally add **custom events** on the grid or by natural language, then use schedule-attached chat for NL add/drop/swap with stable orchestration and integration hardening.

- ▶ Toggle **list / calendar**; show blocks with course code, section, and local time range; mention **custom blocks** if enabled in your build.
- ▶ One NL mutation on the deployed app; metrics and search in schedule chat respect **allowlisted** resolved course context so the model cannot invent schedule rows.
- ▶ If time: walk a **clarification** loop for an ambiguous section or add, with state saved per conversation until answers resume the flow; optionally show section-aware add or swap messaging when sections collide.



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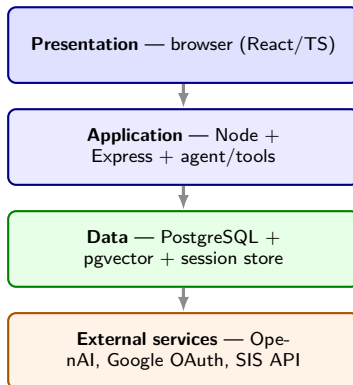
Future work and reflection

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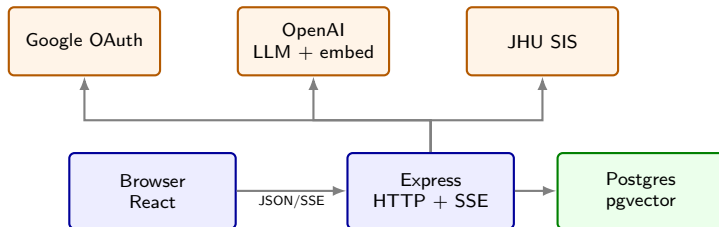
Stack: Logical Layers

Responsibilities

- ▶ **Client:** React + Vite, streaming UI, schedule views
- ▶ **API:** Express routes, auth session, agent endpoint
- ▶ **Data:** Postgres + pgvector, chat/audit persistence
- ▶ **External:** OpenAI, Google OAuth, JHU SIS



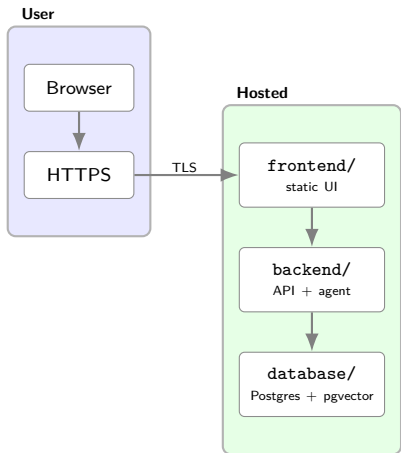
Stack: Components



Integration notes

Session cookie authenticates dashboard routes; `/api/agent` upgrades to SSE for streaming tokens and tool progress.

Deployment



Swap host names as needed: TLS at static frontend/; browser calls backend/; server uses database/ on managed Postgres.

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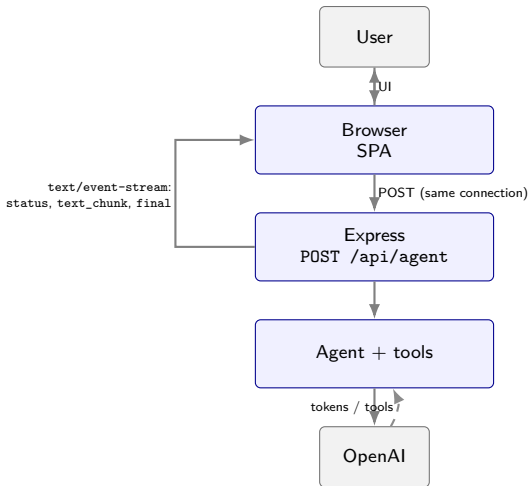
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Runtime: chat request and SSE



User message saved immediately; assistant text on final; partial + aborted metadata on stop.



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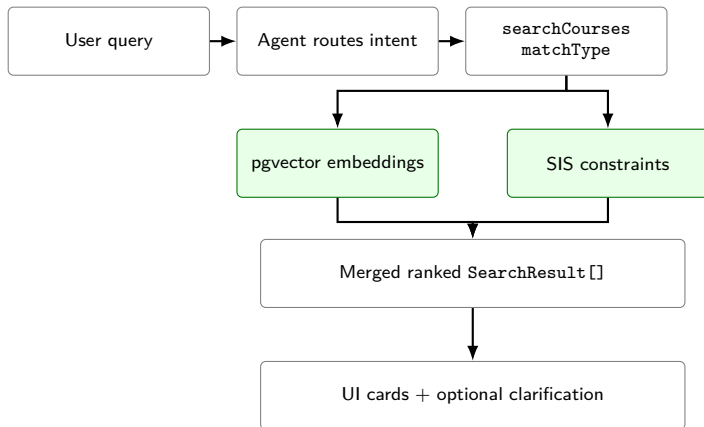
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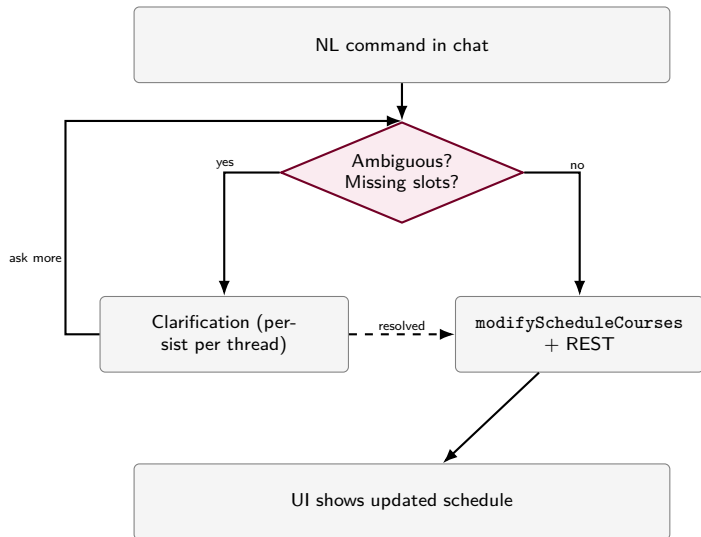
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Data flow: unified course search



One merged search surface exposes semantic, exact, and hybrid retrieval per row via `matchType`.

Control flow: NL schedule edit



Schedule mutations stay blocked until every required clarification slot is confirmed.



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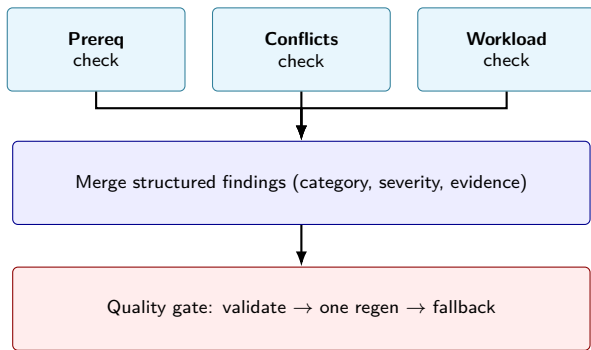
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Parallel Audit Synthesis and Gate



Failure isolation

One failed check still returns partial findings; gate telemetry (gate_pass, gate_fail, ...) where implemented.

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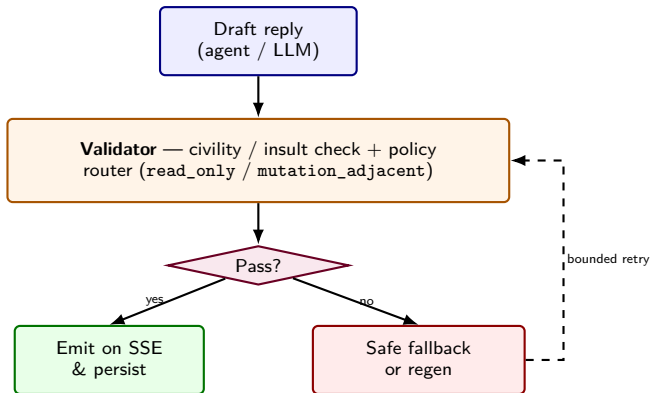
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User-visible text: validator before stream

Assistant text can occasionally include **insulting** or **irrelevant** contents. A **validator** runs on draft output *before* chunks are committed to the SSE stream.



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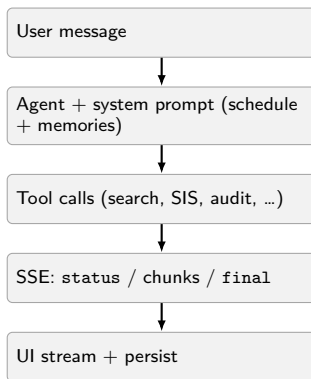
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Agent Loop

Tooling

- ▶ Models: e.g. gpt-4o-mini; embeddings text-embedding-3-small
- ▶ Search, SIS details, eval summary, metrics, audits, schedule mutate
- ▶ Unified searchCourses; clarification state before writes



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Testing, AI eval, observability

Non-AI (backend/ + frontend/)

- ▶ `npm run test` in `backend/` and `frontend/` (Vitest).
- ▶ `frontend/`: `npm run test:e2e` (Playwright).
- ▶ CI and unit/E2E expectations per final delivery rubric.

AI evaluation

- ▶ Golden prompts / structured checks; agent regression paths.
- ▶ Gate telemetry where the audit and output gates are wired for observability.

Coverage

- ▶ `npm run test:coverage` in `backend/` and `frontend/` for line/branch reports (Vitest + v8).
- ▶ Tie spoken numbers to what CI actually runs; note gaps honestly at Q&A.

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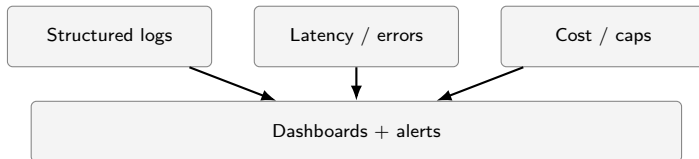
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Ops feedback loop (conceptual)



Final delivery rubric: tie spoken claims to what is actually deployed (logging, rate limits, validation). Mark items as **shipped** vs. **planned** live.

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Closing: Limitations

Where Atlas still falls short

1. **Partial evaluation coverage.** Checks and QA do not yet span every intent or source; stronger semantic analysis across evals, Reddit, RMP, SIS, and chat would support deeper, more comparable scoring.
2. **Smarter agent.** We use gpt-4o-mini for cost and speed; larger or higher-capacity models (more context, stronger reasoning) would help long threads and tool use at higher token cost and latency.
3. **More edge cases.** Fine-tuned flows could cover bespoke plans: weekly templates, explicit time blocks, small constraint tweaks, and advisor-specific modes beyond generic prompts.
4. **Courses vs. sections.** Flows remain course-centric in places; richer section-level control (times, seats, cross-listing) would make adds, swaps, and audits match enrollment reality more closely.

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Closing: Future work & reflection

Future work

Stronger observability dashboards; broader course history and transcript fidelity; tighter prompt and tool contracts; more automated AI evaluation on golden paths; deeper section-aware scheduling where the product roadmap allows.

Reflection

Team coordination across a long build window; using AI tools as development partners; trade-offs between latency and depth, autonomy and safety, and shipping proof over perfection.

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Thank you — Questions

Questions?

Please ask questions about the project, the team, the process, the technology, the future, etc.

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